

[24.11.20] 게임별 인구통계 데이터 (data.ai API)

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1. 개요

🚩 1) 목적

- data.ai API를 활용하여 data.ai 상의 **게임별 인구통계 데이터 추출**
- 수작업으로 진행되었던 기존 게임별 인구통계 데이터 추출 방식의 **자동화**
- 다양한 장르의 게임들을 성별 & 연령별 기준으로 한눈에 보기 쉽게 **시각화**

📋 2) 단계

1. data.ai 사이트에서 개인별 API 키 발급 및 저장
2. 마켓별 & 국가별 & 기간별 전체 게임 리스트 추출
3. 인구통계 데이터를 원하는 게임명 입력
4. 마켓별 인구통계 데이터 기준 통일 및 장르/회사명/출시일/최근업데이트/apk사이즈 병합
5. 게임별 앱 아이콘 주소 추가 병합 및 개별 아이콘 추출 & 저장
6. 추출된 .xlsx 엑셀 파일을 바탕으로 태블로 대시보드 제작

📄 3) 참고

- 아래 코드에서 입력해야 하는 부분 정리
 - **API KEY** : data.ai 에서 개인 API KEY 발급 (admin 승인 필요)
 - **.xlsx 추출** : 저장 경로 설정
 - **아이콘 추출** : 저장 경로 설정
- API 활용 시 유의사항
 - **단계 - 3번** : 마켓별 게임명 다른 경우 & 추출 안되는 경우 있음

- 단계 - 4번 :마켓별 출시일 & 최근업데이트 & apk사이즈 없는 경우 있음
- 단계 - 5번 : 게임별 앱 아이콘 주소는 있으나, 다운로드 안되는 경우 있음

! 데이터 추출 후, 반드시 추출 결과 확인

2. 자동화

단계별 코드

▼ 1) 사전 작업

```
1 # 라이브러리
2 import os
3 import requests
4 import pandas as pd
5 import concurrent.futures
6 from datetime import datetime, timedelta
7
8
9 # API KEY
10 OPEN_API_KEY = 'data.ai에서 발급받은 API KEY 입력'
11
12
13 # 장르번호 및 대장르명 & 세부장르명
14 genre_info = {
15     2001001: (".io", "Hypercasual"),
16     2001002: ("Action", "Hypercasual"),
17     2001003: ("Ball", "Hypercasual"),
18     2001004: ("Music", "Hypercasual"),
19     2001005: ("Puzzle", "Hypercasual"),
20     2001006: ("Racing", "Hypercasual"),
21     2001007: ("Simulation", "Hypercasual"),
22     2001008: ("Shooting", "Hypercasual"),
23     2001009: ("Sports", "Hypercasual"),
24     2001010: ("Other Hypercasual", "Hypercasual"),
25     2001011: ("Escape", "Hypercasual"),
26     2001012: ("Merge HC", "Hypercasual"),
27     2002001: ("Platformer", "Action"),
28     2002002: ("Run & Gun Platformer", "Action"),
29     2002003: ("Fighting", "Action"),
30     2002004: ("MOBA", "Action"),
31     2002005: ("Brawl", "Action"),
32     2002006: ("Shoot'em Up", "Action"),
33     2002007: ("Runner", "Action"),
34     2002008: ("Music & Rhythm", "Action"),
35     2002009: ("Other Action", "Action"),
36     2002010: ("Artillery Shooter", "Action"),
37     2002011: ("Asymmetric Battle", "Action"),
38     2002012: ("Royale Battle", "Action"),
```

```
39 2002013: ("Open World", "Action"),
40 2002014: ("Shark", "Action"),
41 2002015: ("Slicing", "Action"),
42 2002016: ("Stealth", "Action"),
43 2004001: ("M3", "Match"),
44 2004002: ("M3-Meta", "Match"),
45 2004003: ("Blast", "Match"),
46 2004004: ("Line", "Match"),
47 2004005: ("Bubble Shooter", "Match"),
48 2004006: ("Merge 2", "Match"),
49 2004007: ("Other Match", "Match"),
50 2004008: ("Blast-Meta", "Match"),
51 2004009: ("Other-Meta", "Match"),
52 2004010: ("Connect Tiles", "Match"),
53 2004011: ("Merge 3", "Match"),
54 2004012: ("Match 3D", "Match"),
55 2004013: ("Pair", "Match"),
56 2005001: ("Brain Puzzle", "Puzzle"),
57 2005002: ("Hidden Object", "Puzzle"),
58 2005003: ("Jigsaw", "Puzzle"),
59 2005004: ("Physics", "Puzzle"),
60 2005005: ("Sudoku", "Puzzle"),
61 2005006: ("Word", "Puzzle"),
62 2005007: ("Other Puzzle", "Puzzle"),
63 2005008: ("Tile-Matching", "Puzzle"),
64 2005009: ("Detective", "Puzzle"),
65 2005010: ("Hybrid Puzzle", "Puzzle"),
66 2005011: ("Room Escape", "Puzzle"),
67 2005012: ("Block Puzzle", "Puzzle"),
68 2005013: ("Minesweeper", "Puzzle"),
69 2005014: ("Brick-crusher", "Puzzle"),
70 2005015: ("Spot the Difference", "Puzzle"),
71 2005016: ("Maze", "Puzzle"),
72 2005017: ("Nonogram", "Puzzle"),
73 2005018: ("Klotski", "Puzzle"),
74 2005019: ("Adventure Puzzle", "Puzzle"),
75 2008001: ("Battle Royale", "Shooting"),
76 2008002: ("Team Deathmatch", "Shooting"),
77 2008003: ("PvE Shooting", "Shooting"),
78 2008004: ("Cover-fire", "Shooting"),
79 2008005: ("Sniper", "Shooting"),
80 2008006: ("Extraction Shooter", "Shooting"),
81 2009001: ("Avatar Life", "Simulation"),
82 2009002: ("Building", "Simulation"),
83 2009003: ("Build Breed", "Simulation"),
84 2009004: ("Creative Sandbox", "Simulation"),
85 2009005: ("Dating", "Simulation"),
86 2009006: ("Time Management", "Simulation"),
87 2009007: ("Manager", "Simulation"),
88 2009008: ("Farming", "Simulation"),
89 2009009: ("Idle", "Simulation"),
90 2009010: ("Interactive Story", "Simulation"),
91 2009011: ("Home Design", "Simulation"),
92 2009012: ("Dress Up & Lifestyle", "Simulation"),
93 2009013: ("Pet", "Simulation"),
94 2009014: ("Driving", "Simulation"),
95 2009015: ("Other Sim", "Simulation"),
96 2009016: ("Idol Training Sim", "Simulation"),
```

```

97 2009017: ("Other Japanese Sim", "Simulation"),
98 2009018: ("Animal Avatar", "Simulation"),
99 2009019: ("Life Simulator", "Simulation"),
100 2009020: ("Weapon Simulator", "Simulation"),
101 2009021: ("Game Console Emulator", "Simulation"),
102 2011001: ("Auto Chess", "Strategy"),
103 2011002: ("Build-Battle", "Strategy"),
104 2011003: ("CCG-Battle", "Strategy"),
105 2011004: ("Lane-Battle", "Strategy"),
106 2011005: ("4X March-Battle", "Strategy"),
107 2011006: ("Summon-Battle", "Strategy"),
108 2011007: ("Tower Defense Strategy", "Strategy"),
109 2011008: ("Other Strategy", "Strategy"),
110 2011009: ("Legion Battle", "Strategy"),
111 2011010: ("Turn-Based Strategy", "Strategy"),
112 2012001: ("Mafia/Betrayal", "Party"),
113 2012002: ("Party Royale", "Party"),
114 2012003: ("Luck Battle", "Party"),
115 2012005: ("Quiz", "Party"),
116 2012006: ("Other Party", "Party"),
117 2012007: ("Mini Games Collection", "Party"),
118 2012008: ("Guess Something", "Party"),
119 2012009: ("Truth or Dare", "Party"),
120 2013001: ("Card Games Tabletop", "Tabletop"),
121 2013002: ("Solitaire", "Tabletop"),
122 2013003: ("Mahjong Solitaire", "Tabletop"),
123 2013004: ("Hanafuda", "Tabletop"),
124 2013005: ("Go-Stop", "Tabletop"),
125 2013006: ("Othello", "Tabletop"),
126 2013007: ("Board Game", "Tabletop"),
127 2013008: ("Dice", "Tabletop"),
128 2013009: ("Chess", "Tabletop"),
129 2013010: ("Coloring", "Tabletop"),
130 2013012: ("Ludo", "Tabletop"),
131 2007001: ("Open World RPG", "RPG"),
132 2007002: ("Team Battle", "RPG"),
133 2007003: ("Idle RPG", "RPG"),
134 2007004: ("MMORPG", "RPG"),
135 2007005: ("Tactics", "RPG"),
136 2007006: ("Action RPG", "RPG"),
137 2007007: ("Roguelike ARPG", "RPG"),
138 2007008: ("Location RPG", "RPG"),
139 2007009: ("Survival", "RPG"),
140 2007010: ("Puzzle RPG", "RPG"),
141 2007011: ("Other RPG", "RPG"),
142 2007012: ("Tower Defense RPG", "RPG"),
143 2007013: ("Empire Simulation RPG", "RPG")}

```

▼ 2) 전체 게임 리스트 추출

2

```

1 # demographics_product_id 확인 (google)
2 A_google = requests.get(
3     f'https://api.data.ai/v2.0/app_explorer?markets=google-play&revenue_range=10000,99999999999&re
4     headers= {'Authorization': f'Bearer {OPEN_API_KEY}'})
5
6
7 # demographics_product_id 확인 (apple)
8 A_apple = requests.get(

```

```

9         f'https://api.data.ai/v2.0/app_explorer?markets=apple-store&revenue_range=10000,9999999999&re
10         headers= {'Authorization': f'Bearer {OPEN_API_KEY}'}))
11
12
13 # 결과 확인
14 A_google = pd.DataFrame(A_google.json()['data']).sort_values(by= 'product_id', ignore_index= True)
15 A_apple = pd.DataFrame(A_apple.json()['data']).sort_values(by= 'product_id', ignore_index= True)
16 A_list = pd.concat([A_google, A_apple])
17
18 print("* 'app_list'")
19 display(A_list)

```

결과 예시

* 'app_list'							
	product_id	Company Name	App	Company HQ	Category	Initial Release Date	App Store
0	206000000000075	Seadcloud	Baby Connect: Newborn Tracker	US	Health & Fitness	Wed Aug 24 17:00:00 2011	google-play
1	206000000000084	BeFunky	Photo Editor by BeFunky	US	Photo & Video	Sun Mar 13 17:00:00 2011	google-play
2	206000000000160	AudioAddict	DI.FM: Electronic Music Radio	US	Music	Wed Mar 9 16:00:00 2011	google-play
3	206000000000425	Viber Media	Rakuten Viber Messenger	CY	Social	Mon Jul 18 17:00:00 2011	google-play
4	206000000000475	None	HotSchedules	None	Productivity	Mon Aug 16 17:00:00 2010	google-play
...	...	...	...	...	...	...	...
6493	6736572859	None	Feynman AI: Memorize Anything	None	Education	Mon Oct 7 17:00:00 2024	apple-store
6494	6736608790	Digital Melody	Gun Shop Simulator 3D Shooting	PL	Games	Tue Oct 22 17:00:00 2024	apple-store
6495	6736647522	None	Lace - Random Video Chat	None	Social	Wed Oct 9 17:00:00 2024	apple-store
6496	6736899099	NABIR	Mecha: Heart of Steel	None	Games	Wed Oct 30 17:00:00 2024	apple-store
6497	6737254232	None	Feather: Draw in 3D	None	Art & Design	Thu Oct 31 17:00:00 2024	apple-store

3) 희망 게임 리스트 입력

```

3 1 # 게임리스트 입력
2 app_list = ['MONOPOLY GO', 'Royal Match', 'Candy Crush Saga', 'Township', 'Coin Master', 'Marvel Snap',
3         'Gardenscapes', 'Homescapes', 'Fishdom', 'Solitaire Grand Harvest', 'Match Factory',
4         "June's Journey", 'Project Makeover', 'Dice Dreams', 'Domino Dreams', 'Travel Town - Merge
5         'Last War:Survival', 'Whiteout Survival', 'Pokémon GO', 'Brawl Stars', 'Toon Blast',
6         'Clash of Clans', 'Clash Royale', 'Squad Busters', 'Marvel Strike Force', 'Match Masters']
7
8
9 # 데이터 정리
10 product_id_google = A_google[A_google['App'].str.contains('|'.join(app_list), case= False)][['product_id']]
11 product_id_google = product_id_google.drop_duplicates()
12
13 product_id_apple = A_apple[A_apple['App'].str.contains('|'.join(app_list), case= False)][['product_id']]
14 product_id_apple = product_id_apple.drop_duplicates()
15
16
17 # App별 product_id 확인
18 print("* App_List (google)")
19 display(product_id_google)
20 print('\n', '\n')
21
22 print("* App_List (apple)")
23 display(product_id_apple)

```

결과 예시

! 애플에서 미추출된 게임 1개 확인 → 결과 확인 후, product_id_apple에 June's Journey 직접 추가

* App_List (google)			
	product_id	App	App Store
0	20600010081528	Brawl Stars	google-play
1	20600000578686	Candy Crush Saga	google-play
2	20600005325514	Clash Royale	google-play
3	20600001580962	Clash of Clans	google-play
4	20600005233076	Coin Master	google-play
...	...	...	...
22	20600018003005	Squad Busters	google-play
23	20600006900867	Toon Blast	google-play
24	20600001696370	Township	google-play
25	20600014324108	Travel Town - Merge Adventure	google-play
26	20600016782283	Whiteout Survival	google-play

* App_List (apple)			
	product_id	App	App Store
0	1229016807	Brawl Stars	apple-store
1	553834731	Candy Crush Saga	apple-store
2	1053012308	Clash Royale	apple-store
3	529479190	Clash of Clans	apple-store
4	406889139	Coin Master	apple-store
...	...	...	...
21	1668983788	Squad Busters	apple-store
22	1176027022	Toon Blast	apple-store
23	638689075	Township	apple-store
24	1521236603	Travel Town - Merge Adventure	apple-store
25	6443575749	Whiteout Survival	apple-store

4) 게임별 앱 아이콘 확인

```

4
1 # 미포함 게임 추가
2 product_id_apple.loc[len(product_id_apple)] = [1200391796, "June's Journey", "apple-store"]
3
4
5 # 'google_app_icon' 추출 함수
6 def fetch_google_app_icon(app_id):
7     google_app_icon = requests.get(
8         f"https://api.data.ai/v1.3/apps/google-play/app/{app_id}/details",
9         headers= {"Authorization": f"Bearer {OPEN_API_KEY}"}) .json()
10    google_app_icon_df = pd.DataFrame(columns= ['product_id', 'icon_web', 'first_release_date', 'current_release_date'])
11    google_app_icon_df.loc[len(google_app_icon)] = [google_app_icon['product']['product_id'], google_app_icon['product']['icon_web'], google_app_icon['product']['first_release_date'], google_app_icon['product']['current_release_date']]
12    return google_app_icon_df.reset_index(drop= True)
13
14
15 # 'apple_app_icon' 추출 함수
16 def fetch_apple_app_icon(app_id):
17     apple_app_icon = requests.get(
18         f"https://api.data.ai/v1.3/apps/ios/app/{app_id}/details",
19         headers= {"Authorization": f"Bearer {OPEN_API_KEY}"}) .json()
20     apple_app_icon_df = pd.DataFrame(columns= ['product_id', 'icon_web', 'first_release_date', 'current_release_date'])
21     apple_app_icon_df.loc[len(apple_app_icon)] = [apple_app_icon['product']['product_code'], apple_app_icon['product']['icon_web'], apple_app_icon['product']['first_release_date'], apple_app_icon['product']['current_release_date']]
22     return apple_app_icon_df.reset_index(drop= True)
23
24

```

결과 예시

[illegible]

* App_List (apple)									
	product_id	App	App Store	first_release_date	current_release_date	last_size	icon_web		
0	1229016807	Brawl Stars	apple-store	2017-06-15	2024-10-29	1228	https://static-saa-cdn.net/img/ios/1229016807/13e57a84fef49645df84382b02bdc267_w80		
1	553834731	Candy Crush Saga	apple-store	2012-11-14	2024-11-12	385	https://static-saa-cdn.net/img/ios/553834731/4e4524e65834f1c16d273717d5a77064e_w80		
2	1053012308	Clash Royale	apple-store	2016-01-04	2024-11-15	764	https://static-saa-cdn.net/img/ios/1053012308/c5644528d4aaefa1545362d1a2d1495_w80		
3	529479190	Clash of Clans	apple-store	2012-06-14	2024-11-05	392	https://static-saa-cdn.net/img/ios/529479190/b42431e61d52a49572276a88ac135f56_w80		
4	406889139	Coin Master	apple-store	2010-12-11	2024-11-11	267	https://static-saa-cdn.net/img/ios/406889139/373dc621cf9bc293887b65c166dd0c12_w80		
...	...	...	...	...	...	...	...		
22	1176027022	Toon Blast	apple-store	2017-01-11	2024-11-13	412	https://static-saa-cdn.net/img/ios/1176027022/40860dfe9f17644c28cae89aa123f8f_w80		
23	638689075	Township	apple-store	2013-10-24	2024-11-11	235	https://static-saa-cdn.net/img/ios/638689075/fce23f69aa43e643db57cf08756c396e_w80		
24	1521236603	Travel Town - Merge Adventure	apple-store	2020-07-10	2024-11-11	287	https://static-saa-cdn.net/img/ios/1521236603/9ee25af6a107c6436e54211b08b1e507_w80		
25	6443575749	Whiteout Survival	apple-store	2022-12-02	2024-10-19	772	https://static-saa-cdn.net/img/ios/6443575749/b756746aa0a004d530d5277145817b7c_w80		
26	1200391796	June's Journey	apple-store	2017-04-21	2024-11-11	533	https://static-saa-cdn.net/img/ios/1200391796/1005648ed457198db3e02099cd07a229_w80		

5) 게임별 추가 정보 추출

```

5
1 # 장르 & 서브장르 분리
2 def set_genre_info(df, genre_id):
3     genre_name, subgenre = genre_info.get(genre_id, ("Unknown", "Unknown"))
4     df['genre_name'] = genre_name
5     df['subgenre'] = subgenre
6     return df
7
8
9 # 'genre_id'별 전체 앱 추출
10 def fetch_and_process_genre_data(genre_id_def):
11     response = requests.get(
12         f'https://api.data.ai/v2.0/app_explorer?markets=apple-store,google-play&genre_ids={genre_id_def}',
13         headers={'Authorization': f'Bearer {OPEN_API_KEY}'})
14
15     data = pd.DataFrame(response.json()['data'])
16     return set_genre_info(data, genre_id_def)
17
18
19 # 전체 앱 중 기준에 맞는 앱 선택
20 def apply_genre_info_to_all(genre_info_def):
21     final_df = pd.DataFrame()
22
23     with concurrent.futures.ThreadPoolExecutor(max_workers= 10) as executor:
24         results = list(executor.map(fetch_and_process_genre_data, genre_info_def))
25
26     for result in results:
27         final_df = pd.concat([final_df, result], ignore_index= True)
28     return final_df
29
30
31 # 칼럼 정리
32 final_result_app_list = apply_genre_info_to_all(genre_info)
33 final_result_app_list['product_id'] = final_result_app_list['product_id'].astype(float).astype(int).as
34 final_result_app_list = final_result_app_list.rename(columns= {'subgenre': 'genre', 'genre_name': 'sub
35 final_result_app_list = final_result_app_list[['product_id', 'Company Name', 'App', 'genre', 'subgenre
36
37
38 # app_list 통합 및 장르/서브장르 병합
39 app_list_result = pd.concat([app_list_google, app_list_apple])
40 app_list_result = app_list_result.drop_duplicates()
41 final_result_app_list['product_id'] = final_result_app_list['product_id'].astype(float).astype(int).as
42
43 app_list_final_result = app_list_result.merge(final_result_app_list[['product_id', 'App Store', 'genre
44 app_list_final_result = app_list_final_result[['product_id', 'App', 'Company Name', 'App Store', 'genr
45
46

```

```

1  # product_id 리스트 생성
2  B_google = app_list_google['product_id']
3  demographics_google = pd.DataFrame()
4
5  end_date = datetime.now()
6  start_date = end_date - pd.DateOffset(months= 12)
7  start_dates = pd.date_range(start= start_date, end= end_date - pd.DateOffset(months= 1), freq= 'MS').
8
9
10 # 병렬처리로 demographics 추출
11 with concurrent.futures.ThreadPoolExecutor(max_workers= 10) as executor:
12     futures = []
13
14     for product_id in B_google:
15         for start_date in start_dates:
16             futures.append(executor.submit(
17                 requests.get,
18                 f"https://api.data.ai/v1.2/intelligence/apps/all-android/app/{product_id}/demographics
19                 headers={"Authorization": f"Bearer {OPEN_API_KEY}"})
20
21 for future in concurrent.futures.as_completed(futures):
22     C_google = future.result().json()
23     D_google = pd.concat([
24         pd.DataFrame({
25             'product_id': [C_google['product_id']],
26             'product_name': [C_google['product_name']],
27             'company_name': [C_google['company_name']],
28             'unified_product_name': [C_google['unified_product_name']]},
29         pd.DataFrame(C_google['list']), axis= 1)
30
31     demographics_google = pd.concat([demographics_google, D_google], ignore_index= True)
32
33
34 # 칼럼 정리
35 demographics_google['date'] = demographics_google['start_date'].str.slice(2, 7)
36 demographics_google['date'] = pd.to_datetime(demographics_google['date'], format= '%y-%m').dt.strftime
37 demographics_google['store'] = 'google-play'
38 demographics_google = demographics_google.drop(['start_date', 'end_date'], axis= 1)
39 demographics_google = demographics_google.replace('n/a', pd.NA)

```

```

40 demographics_google = demographics_google.dropna()
41 demographics_google = demographics_google.sort_values(by= ['product_name', 'date'], ignore_index= True)
42 demographics_google['age_index_25_44'] = demographics_google['age_index_25_34'] + demographics_google[
43 demographics_google['age_percent_25_44'] = demographics_google['age_percent_25_34'] + demographics_goc
44
45 del demographics_google['age_index_25_34']
46 del demographics_google['age_index_35_44']
47 del demographics_google['age_percent_25_34']
48 del demographics_google['age_percent_35_44']
49
50
51 # 아이콘 주소 병합
52 demographics_google = pd.merge(demographics_google, app_list_google[['product_id', 'icon_web']], on= '
53
54
55 # 비율을 가중치로 계산한 평균 연령대 추정
56 demographics_google['estimated_age'] = (
57     20 * demographics_google['age_percent_18_24'] / 100 +
58     34.5 * demographics_google['age_percent_25_44'] / 100 +
59     52.5 * demographics_google['age_percent_45_plus'] / 100)
60 demographics_google['estimated_age'].astype(float)
61
62
63 # 기존 결과 DF와 병합
64 app_list_final_result_merge = app_list_final_result[['product_id', 'App Store', 'genre', 'subgenre', '
65 app_list_final_result_merge = app_list_final_result_merge.rename(columns= {'App Store': 'store'})
66
67 demographics_google = pd.merge(demographics_google, app_list_final_result_merge, on= ['product_id', 's
68
69
70 # 칼럼 정리
71 demographics_google = demographics_google[['product_id', 'product_name', 'company_name', 'genre', 'sub
72     'age_index_18_24', 'age_percent_18_24', 'age_index_25_44',
73     'gender_index_female', 'gender_percent_female', 'gender_inc
74
75
76 # 결과 확인
77 print("* demographics (google)")
78 display(demographics_google)
79
80
81 # .xlsx 추출
82 demographics_google.to_excel('내 컴퓨터의 저장공간 위치 입력', index= False)

```

결과 예시

	product_id	product_name	company_name	genre	subgenre	first_release_date	current_release_date	current_app_size	estimated_age	age_index_18_24	age_percent_18_24	age_index_25_44	age_percent_25_44	age_index_45_plus	age_percent_45_plus	gender_index
0	20600010081528	Brave Stars	Supercell	Action	Brave	2018-06-26	2024-10-25	368	33.51	199	40.24	175	32.85	63	26.91	
1	20600010081528	Brave Stars	Supercell	Action	Brave	2018-06-26	2024-10-25	368	33.82	193	38.99	178	33.39	64	27.62	
2	20600010081528	Brave Stars	Supercell	Action	Brave	2018-06-26	2024-10-25	368	34.26	185	37.27	182	34.05	67	28.68	
3	20600010081528	Brave Stars	Supercell	Action	Brave	2018-06-26	2024-10-25	368	34.29	188	37.93	174	32.69	68	29.38	
4	20600010081528	Brave Stars	Supercell	Action	Brave	2018-06-26	2024-10-25	368	34.14	182	38.76	171	33.03	68	29.21	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
287	20600016782283	Whitewood Survival	Century Games	Strategy	4X March-Battle	2022-06-28	2024-10-14	0	36.44	138	27.96	207	38.73	78	33.31	
288	20600016782283	Whitewood Survival	Century Games	Strategy	4X March-Battle	2022-06-28	2024-10-14	0	37.89	121	24.42	198	37.06	90	38.52	
289	20600016782283	Whitewood Survival	Century Games	Strategy	4X March-Battle	2022-06-28	2024-10-14	0	37.34	124	25.00	210	39.10	84	35.90	
290	20600016782283	Whitewood Survival	Century Games	Strategy	4X March-Battle	2022-06-28	2024-10-14	0	37.21	127	25.75	206	38.45	83	35.80	
291	20600016782283	Whitewood Survival	Century Games	Strategy	4X March-Battle	2022-06-28	2024-10-14	0	37.78	116	23.46	211	39.40	87	37.14	

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	product_id	product_name	company_name	genre	subgenre	release_dttm	release_apk	estimated_size	index_1	index_2	percent_1	percent_2	index_3	percent_3	index_4	percent_4	index_5	percent_5	date	store	icon_web	
2	20600010C	Brawl Star	Supercell	Action	Brawl	2018-06-2 2024-10-2	368	33.5911	199	40.2407	175	32.9484	63	26.911	57	27.0578	137	72.9422	2023-12	google	pl=https://stat	
3	20600010C	Brawl Star	Supercell	Action	Brawl	2018-06-2 2024-10-2	368	33.8164	193	38.9946	178	33.3884	64	27.617	62	29.3421	132	70.0579	2024-01	google	pl=https://stat	
4	20600010C	Brawl Star	Supercell	Action	Brawl	2018-06-2 2024-10-2	368	34.25725	185	37.2747	182	34.0468	67	28.6784	63	29.9459	131	70.0541	2024-02	google	pl=https://stat	
5	20600010C	Brawl Star	Supercell	Action	Brawl	2018-06-2 2024-10-2	368	34.28829	188	37.9333	174	32.6852	68	29.3814	63	29.7691	132	70.2309	2024-03	google	pl=https://stat	
6	20600010C	Brawl Star	Supercell	Action	Brawl	2018-06-2 2024-10-2	368	34.13748	192	38.7623	171	32.0268	68	29.211	61	28.8676	133	71.1324	2024-04	google	pl=https://stat	
7	20600010C	Brawl Star	Supercell	Action	Brawl	2018-06-2 2024-10-2	368	34.02706	194	39.1328	171	31.971	67	28.8962	61	28.9685	133	71.0315	2024-05	google	pl=https://stat	
8	20600010C	Brawl Star	Supercell	Action	Brawl	2018-06-2 2024-10-2	368	34.32343	188	37.9211	173	32.5723	69	29.5666	63	29.936	131	70.064	2024-06	google	pl=https://stat	
9	20600010C	Brawl Star	Supercell	Action	Brawl	2018-06-2 2024-10-2	368	34.78685	176	35.6083	181	34.1136	71	30.2781	63	29.6662	132	70.3338	2024-07	google	pl=https://stat	
10	20600010C	Brawl Star	Supercell	Action	Brawl	2018-06-2 2024-10-2	368	34.83971	179	36.1229	176	32.8908	72	30.9863	65	30.5999	130	69.4001	2024-08	google	pl=https://stat	
11	20600010C	Brawl Star	Supercell	Action	Brawl	2018-06-2 2024-10-2	368	35.27787	173	34.9822	173	32.5162	76	32.5016	67	31.6149	128	68.3851	2024-09	google	pl=https://stat	
12	20600010C	Brawl Star	Supercell	Action	Brawl	2018-06-2 2024-10-2	368	35.10115	177	35.707	172	32.1893	75	32.1037	67	31.6119	128	68.3881	2024-10	google	pl=https://stat	
13	206000005	Candy Cr.	King	Match	M3	2012-11-1 2024-11-0	0	42.31269	57	11.5861	190	35.6748	123	52.7371	131	61.7325	71	38.2675	2023-12	google	pl=https://stat	
14	206000005	Candy Cr.	King	Match	M3	2012-11-1 2024-11-0	0	42.38057	54	10.9844	194	36.3864	123	52.6293	130	60.9187	73	39.0813	2024-01	google	pl=https://stat	
15	206000005	Candy Cr.	King	Match	M3	2012-11-1 2024-11-0	0	42.41897	53	10.8183	194	36.4727	123	52.709	132	61.9825	71	38.0175	2024-02	google	pl=https://stat	
16	206000005	Candy Cr.	King	Match	M3	2012-11-1 2024-11-0	0	42.42536	54	11.0299	193	36.0551	124	52.915	129	60.7307	73	39.2693	2024-03	google	pl=https://stat	
17	206000005	Candy Cr.	King	Match	M3	2012-11-1 2024-11-0	0	42.18822	57	11.6227	194	36.2945	122	52.0785	128	60.3501	74	39.6499	2024-04	google	pl=https://stat	
18	206000005	Candy Cr.	King	Match	M3	2012-11-1 2024-11-0	0	42.16091	59	11.913	192	35.5290	122	52.1572	130	61.1933	72	38.8087	2024-05	google	pl=https://stat	
19	206000005	Candy Cr.	King	Match	M3	2012-11-1 2024-11-0	0	42.20156	59	11.967	190	35.6065	122	52.4265	130	61.2319	72	38.7681	2024-06	google	pl=https://stat	
20	206000005	Candy Cr.	King	Match	M3	2012-11-1 2024-11-0	0	42.33255	58	11.7868	188	35.2038	124	53.0093	132	61.8952	71	38.1048	2024-07	google	pl=https://stat	
21	206000005	Candy Cr.	King	Match	M3	2012-11-1 2024-11-0	0	41.98702	61	12.3889	192	36.0366	120	51.5745	131	61.6937	72	38.3063	2024-08	google	pl=https://stat	
22	206000005	Candy Cr.	King	Match	M3	2012-11-1 2024-11-0	0	42.14028	59	12.0013	191	35.8853	122	52.1135	129	60.7507	73	39.2493	2024-09	google	pl=https://stat	
23	206000005	Candy Cr.	King	Match	M3	2012-11-1 2024-11-0	0	42.42303	57	11.6118	187	35.0174	125	53.3708	131	61.7286	71	38.2714	2024-10	google	pl=https://stat	
24	206000005	Clash Roy.	Supercell	Strategy	Summon	2016-02-1 2024-11-1	151	32.10875	218	44.012	181	33.8186	52	22.1694	45	21.458	147	78.542	2023-12	google	pl=https://stat	
25	206000005	Clash Roy.	Supercell	Strategy	Summon	2016-02-1 2024-11-1	151	32.40001	211	42.523	186	34.889	52	22.588	44	21.0149	148	78.9851	2024-01	google	pl=https://stat	
26	206000005	Clash Roy.	Supercell	Strategy	Summon	2016-02-1 2024-11-1	151	32.98699	202	40.8388	185	34.6689	57	24.4923	48	22.7218	145	77.2782	2024-02	google	pl=https://stat	
27	206000005	Clash Roy.	Supercell	Strategy	Summon	2016-02-1 2024-11-1	151	32.46031	213	42.8791	181	33.911	54	23.2099	46	21.8551	146	78.1449	2024-03	google	pl=https://stat	
28	206000005	Clash Roy.	Supercell	Strategy	Summon	2016-02-1 2024-11-1	151	32.19889	221	44.5444	172	32.3513	54	23.1044	46	21.7918	147	78.2082	2024-04	google	pl=https://stat	
29	206000005	Clash Roy.	Supercell	Strategy	Summon	2016-02-1 2024-11-1	151	32.35234	219	44.2078	171	32.1115	55	23.6806	46	21.8835	146	78.1165	2024-05	google	pl=https://stat	
30	206000005	Clash Roy.	Supercell	Strategy	Summon	2016-02-1 2024-11-1	151	32.64199	211	42.4882	179	33.6075	56	23.9043	46	21.9777	146	78.0223	2024-06	google	pl=https://stat	
31	206000005	Clash Roy.	Supercell	Strategy	Summon	2016-02-1 2024-11-1	151	32.90889	201	40.5505	191	35.6233	55	23.8262	47	22.159	146	77.841	2024-07	google	pl=https://stat	
32	206000005	Clash Roy.	Supercell	Strategy	Summon	2016-02-1 2024-11-1	151	33.25788	198	39.8908	186	34.8756	59	25.2336	49	23.082	144	76.918	2024-08	google	pl=https://stat	
33	206000005	Clash Roy.	Supercell	Strategy	Summon	2016-02-1 2024-11-1	151	32.81853	201	41.7817	181	33.9023	57	24.316	48	22.6373	145	77.3627	2024-09	google	pl=https://stat	
34	206000005	Clash Roy.	Supercell	Strategy	Summon	2016-02-1 2024-11-1	151	32.59939	214	42.0798	175	32.776	56	24.1442	49	23.3966	144	76.6024	2024-10	google	pl=https://stat	
35	206000011	Clash of C	Supercell	Strategy	Build-Batt	2013-09-3 2024-11-0	275	33.29619	199	40.197	183	34.1099	60	25.6931	44	21.0574	148	78.9426	2023-12	google	pl=https://stat	
36	206000001	Clash of C	Supercell	Strategy	Build-Batt	2013-09-3 2024-11-0	275	33.35275	197	39.6442	186	34.7938	59	25.562	47	22.1034	146	77.8966	2024-01	google	pl=https://stat	
37	206000001	Clash of C	Supercell	Strategy	Build-Batt	2013-09-3 2024-11-0	275	33.6013	193	38.9094	185	34.7394	61	26.3511	49	23.3641	144	76.6359	2024-02	google	pl=https://stat	
38	206000001	Clash of C	Supercell	Strategy	Build-Batt	2013-09-3 2024-11-0	275	33.3596	199	40.1525	181	33.8383	61	26.0093	46	21.8593	146	78.1407	2024-03	google	pl=https://stat	
39	206000001	Clash of C	Supercell	Strategy	Build-Batt	2013-09-3 2024-11-0	275	34.3804	200	40.4267	176	33.0849	62	26.4894	48	22.7563	145	77.2437	2024-04	google	pl=https://stat	
40	206000001	Clash of C	Supercell	Strategy	Build-Batt	2013-09-3 2024-11-0	275	33.4483	197	39.7796	181	33.8257	61	26.3947	47	22.2711	146	77.7289	2024-05	google	pl=https://stat	
41	206000001	Clash of C	Supercell	Strategy	Build-Batt	2013-09-3 2024-11-0	275	33.81161	192	38.645	182	34.0487	64	27.3063	49	23.1026	144	76.8974	2024-06	google	pl=https://stat	
42	206000001	Clash of C	Supercell	Strategy	Build-Batt	2013-09-3 2024-11-0	275	34.38425	180	36.2662	188	35.1624	67	28.5714	49	23.1892	144	76.8108	2024-07	google	pl=https://stat	
43	206000001	Clash of C	Supercell	Strategy	Build-Batt	2013-09-3 2024-11-0	275	34.09062	185	37.4111	185	34.7262	65	27.8626	48	22.8327	145	77.1673	2024-08	google	pl=https://stat	
44	206000001	Clash of C	Supercell	Strategy	Build-Batt	2013-09-3 2024-11-0	275	34.1856	180	37.9243	180	33.5891	66	28.4866	48	22.5979	145	77.4021	2024-09	google	pl=https://stat	
45	206000001	Clash of C	Supercell	Strategy	Build-Batt	2013-09-3 2024-11-0	275	34.02334	189	38.0555	182	33.9368	65	28.0077	48	22.4971	145	77.5029	2024-10	google	pl=https://stat	
46	206000005	Coin Mast Moon	Act Party	Luck	Battli	2016-01-2 2024-11-1	0	41.56821	54	11.0593	218	40.764	113	48.1767	113	52.9992	88	47.0008	2023-12	google	pl=https://stat	
47	206000005	Coin Mast Moon	Act Party	Luck	Battli	2016-01-2 2024-11-1	0	41.52037	56	11.2954	217	40.6035	112	48.1011	111	52.0502	90	47.9498	2024-01	google	pl=https://stat	
48	206000005	Coin Mast Moon	Act Party	Luck	Battli	2016-01-2 2024-11-1	0	41.85839	56	11.4396	206	38.4652	117	50.0952	114	53.5842	87	46.4158	2024-02	google	pl=https://stat	
49	206000005	Coin Mast Moon	Act Party	Luck	Battli	2016-01-2 2024-11-1	0	41.65073	56	11.4171	212	39.6598	114	48.9232	110	51.6757	90	48.3243	2024-03	google	pl=https://stat	

7) 게임별 인구통계 데이터 추출 (애플)

7

```

1 # product_id 리스트 생성
2 B_apple = app_list_apple['product_id']
3 demographics_apple = pd.DataFrame()
4
5 end_date = datetime.now()
6 start_date = end_date - pd.DateOffset(months= 12)
7 start_dates = pd.date_range(start= start_date, end= end_date - pd.DateOffset(months= 1), freq= 'MS').as
8
9
10 # 병렬처리로 demographics 추출
11 with concurrent.futures.ThreadPoolExecutor(max_workers= 10) as executor:
12     futures = []
13
14     for product_id in B_apple:
15         for start_date in start_dates:
16             futures.append(executor.submit(
17                 requests.get,
18                 f"https://api.data.ai/v1.2/intelligence/apps/ios/app/{product_id}/demographics?countri
19                 headers={"Authorization": f"Bearer {OPEN_API_KEY}"})
20
21     for future in concurrent.futures.as_completed(futures):
22         C_apple = future.result().json()
23         D_apple = pd.concat([
24             pd.DataFrame({
25                 'product_id': [C_apple['product_id']],
26                 'product_name': [C_apple['product_name']],
27                 'company_name': [C_apple['company_name']],
28                 'unified_product_name': [C_apple['unified_product_name']]},
29             pd.DataFrame(C_apple['list']), axis= 1)
30
31         demographics_apple = pd.concat([demographics_apple, D_apple], ignore_index= True)
32
33
34 # 칼럼 정리/
35 demographics_apple['date'] = demographics_apple['start_date'].str.slice(2, 7)
36 demographics_apple['date'] = pd.to_datetime(demographics_apple['date'], format= '%y-%m').dt.strftime('%

```

```

37 demographics_apple['store'] = 'apple-store'
38 demographics_apple = demographics_apple.drop(['start_date', 'end_date'], axis= 1)
39 demographics_apple = demographics_apple.replace('n/a', pd.NA)
40 demographics_apple = demographics_apple.dropna()
41 demographics_apple = demographics_apple.sort_values(by= ['product_name', 'date'], ignore_index= True)
42
43
44 # 아이콘 주소 병합
45 demographics_apple = pd.merge(demographics_apple, app_list_apple[['product_id', 'icon_web']], on= 'product_id')
46
47
48 # 비율을 가중치로 계산한 평균 연령대 추정
49 demographics_apple['estimated_age'] = (
50     20 * demographics_apple['age_percent_16_24'] / 100 +
51     34.5 * demographics_apple['age_percent_25_44'] / 100 +
52     52.5 * demographics_apple['age_percent_45_plus'] / 100)
53 demographics_apple['estimated_age'].astype(float)
54
55
56 # 기존 결과 DF와 병합
57 app_list_final_result_merge = app_list_final_result[['product_id', 'App Store', 'genre', 'subgenre', 'icon_web']]
58 app_list_final_result_merge = app_list_final_result_merge.rename(columns= {'App Store': 'store'})
59
60 demographics_apple = pd.merge(demographics_apple, app_list_final_result_merge, on= ['product_id', 'store'])
61
62
63 # 칼럼 정리
64 demographics_apple = demographics_apple[['product_id', 'product_name', 'company_name', 'genre', 'subgenre', 'icon_web',
65     'age_index_16_24', 'age_percent_16_24', 'age_index_25_44', 'age_percent_25_44', 'age_index_45_plus', 'age_percent_45_plus',
66     'gender_index_female', 'gender_percent_female', 'gender_index_male', 'gender_percent_male']]
67
68
69 # 결과 확인
70 print("* demographics (apple)")
71 display(demographics_apple)

```

결과 예시

! 애플은 구글에 비해 누락된 데이터가 많고, 부정확한 수치도 존재함. 반드시 유의!

demographics (apple)															
	product_id	product_name	company_name	genre	subgenre	first_release_date	current_release_date	current_app_size	estimated_age	age_index_16_24	age_percent_16_24	age_index_25_44	age_percent_25_44	age_index_45_plus	age_percent_45_plus
0	1229016807	Brawl Stars	Supercell	Action	Brawl	2017-06-15	2024-10-29	1228	28.20	184	56.68	74	32.85	41	10.67
1	1229016807	Brawl Stars	Supercell	Action	Brawl	2017-06-15	2024-10-29	1228	28.30	182	56.14	75	33.09	42	10.77
2	1229016807	Brawl Stars	Supercell	Action	Brawl	2017-06-15	2024-10-29	1228	28.39	182	56.27	73	32.34	44	11.39
3	1229016807	Brawl Stars	Supercell	Action	Brawl	2017-06-15	2024-10-29	1228	28.18	186	57.35	72	31.58	43	11.07
4	1229016807	Brawl Stars	Supercell	Action	Brawl	2017-06-15	2024-10-29	1228	28.19	185	57.00	73	32.14	42	10.85
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
193	638689075	Township	Playrix	Simulation	Farming	2013-10-24	2024-11-11	235	35.95	84	25.97	102	45.05	113	28.98
194	638689075	Township	Playrix	Simulation	Farming	2013-10-24	2024-11-11	235	36.39	82	25.52	99	43.40	122	31.08
195	638689075	Township	Playrix	Simulation	Farming	2013-10-24	2024-11-11	235	36.64	77	23.78	103	45.15	122	31.67
196	638689075	Township	Playrix	Simulation	Farming	2013-10-24	2024-11-11	235	36.69	73	22.63	107	46.97	119	30.40
197	638689075	Township	Playrix	Simulation	Farming	2013-10-24	2024-11-11	235	36.16	77	23.99	108	47.45	112	28.56

product_id	brand	name	category	release_date	release_date_iso	release_date_iso	release_date_iso	release_date_iso	release_date_iso	release_date_iso	release_date_iso	release_date_iso	release_date_iso	release_date_iso	release_date_iso	release_date_iso	release_date_iso	release_date_iso	release_date_iso	release_date_iso	release_date_iso	release_date_iso
12290168	Bravil	Star Supercell	Action	2017-06-1	2024-10-2	1228	28.2023	184	56.6776	74	32.6523	41	10.6699	46	26.4338	170	73.5662	2023-12	apple-store	https://sta		
12290168	Bravil	Star Supercell	Action	2017-06-1	2024-10-2	1228	28.29003	182	56.1436	75	33.0085	42	10.7714	46	26.6438	170	73.5662	2024-01	apple-store	https://sta		
12290168	Bravil	Star Supercell	Action	2017-06-1	2024-10-2	1228	28.39224	182	56.2654	73	32.3417	44	11.3929	51	29.3006	164	70.6994	2024-02	apple-store	https://sta		
12290168	Bravil	Star Supercell	Action	2017-06-1	2024-10-2	1228	28.17633	186	57.3496	72	31.5836	43	11.0668	49	27.9801	167	72.0199	2024-03	apple-store	https://sta		
12290168	Bravil	Star Supercell	Action	2017-06-1	2024-10-2	1228	28.18819	185	57.0033	73	32.143	42	10.8537	47	27.1113	169	72.8887	2024-04	apple-store	https://sta		
12290168	Bravil	Star Supercell	Action	2017-06-1	2024-10-2	1228	27.98175	189	58.324	71	31.3497	40	10.2363	46	26.2431	171	73.7569	2024-05	apple-store	https://sta		
12290168	Bravil	Star Supercell	Action	2017-06-1	2024-10-2	1228	28.23942	183	56.5657	74	32.6488	42	10.7856	48	27.8337	167	72.1663	2024-06	apple-store	https://sta		
12290168	Bravil	Star Supercell	Action	2017-06-1	2024-10-2	1228	28.63679	177	54.5709	77	34.0426	44	11.3865	51	29.2427	164	70.7573	2024-07	apple-store	https://sta		
12290168	Bravil	Star Supercell	Action	2017-06-1	2024-10-2	1228	28.40775	180	55.5586	76	33.5317	42	10.9097	52	29.7249	163	70.2751	2024-08	apple-store	https://sta		
12290168	Bravil	Star Supercell	Action	2017-06-1	2024-10-2	1228	28.42023	181	55.7387	75	33.1372	43	11.1241	52	29.8331	163	70.1669	2024-09	apple-store	https://sta		
12290168	Bravil	Star Supercell	Action	2017-06-1	2024-10-2	1228	28.37822	182	56.1689	74	32.5938	44	11.2373	52	29.6818	163	70.3182	2024-10	apple-store	https://sta		
55384731	Candy Cr.	King	Match	2012-11-1	2024-11-1	385	40.00603	33	10.2211	116	50.9862	152	38.7927	136	77.87	51	22.13	2023-12	apple-store	https://sta		
55384731	Candy Cr.	King	Match	2012-11-1	2024-11-1	385	40.00603	32	9.9887	117	51.3647	151	38.6466	137	76.2534	50	21.7466	2024-01	apple-store	https://sta		
55384731	Candy Cr.	King	Match	2012-11-1	2024-11-1	385	40.24471	31	9.7151	115	50.5438	156	39.7411	136	77.867	51	22.133	2024-02	apple-store	https://sta		
55384731	Candy Cr.	King	Match	2012-11-1	2024-11-1	385	40.10882	32	10.0682	115	50.6612	154	39.2706	136	77.7001	51	22.2999	2024-03	apple-store	https://sta		
55384731	Candy Cr.	King	Match	2012-11-1	2024-11-1	385	39.87317	34	10.5771	116	51.0515	150	38.3714	135	77.1455	53	22.8545	2024-04	apple-store	https://sta		
55384731	Candy Cr.	King	Match	2012-11-1	2024-11-1	385	39.93089	33	10.3265	116	51.1833	151	38.4902	136	77.9561	51	22.0419	2024-05	apple-store	https://sta		
55384731	Candy Cr.	King	Match	2012-11-1	2024-11-1	385	40.2208	32	9.9792	114	50.1998	156	39.821	136	77.9337	51	22.0663	2024-06	apple-store	https://sta		
55384731	Candy Cr.	King	Match	2012-11-1	2024-11-1	385	40.06106	32	9.9503	116	51.1394	152	38.9103	136	77.6571	51	22.3429	2024-07	apple-store	https://sta		
55384731	Candy Cr.	King	Match	2012-11-1	2024-11-1	385	40.0841	34	10.6254	113	49.7925	155	39.5821	137	78.0729	50	21.9271	2024-08	apple-store	https://sta		
55384731	Candy Cr.	King	Match	2012-11-1	2024-11-1	385	40.02341	33	10.3635	115	50.6028	153	39.0338	135	76.9267	53	23.0733	2024-09	apple-store	https://sta		
55384731	Candy Cr.	King	Match	2012-11-1	2024-11-1	385	40.13891	33	10.387	114	50.099	155	39.814	134	76.5251	54	23.4749	2024-10	apple-store	https://sta		
10530123	Clash Roy	Supercell	Strategy	2016-01-0	2024-11-1	764	24.21729	246	75.8557	46	20.1645	15	3.9798	25	14.6704	198	85.3296	2023-12	apple-store	https://sta		
10530123	Clash Roy	Supercell	Strategy	2016-01-0	2024-11-1	764	24.47069	242	74.5128	48	21.1814	16	4.3058	27	15.381	196	84.6197	2024-01	apple-store	https://sta		
10530123	Clash Roy	Supercell	Strategy	2016-01-0	2024-11-1	764	24.57938	241	74.3151	47	20.9342	18	4.7506	28	16.2103	194	83.7897	2024-02	apple-store	https://sta		
10530123	Clash Roy	Supercell	Strategy	2016-01-0	2024-11-1	764	24.36591	244	75.2379	46	20.454	16	4.308	28	16.2249	194	83.7751	2024-03	apple-store	https://sta		
10530123	Clash Roy	Supercell	Strategy	2016-01-0	2024-11-1	764	24.12294	248	76.3763	45	19.7487	15	3.875	26	14.9087	197	85.0913	2024-04	apple-store	https://sta		
10530123	Clash Roy	Supercell	Strategy	2016-01-0	2024-11-1	764	24.30518	244	75.2875	47	20.7021	15	4.0104	26	15.0785	197	84.9215	2024-05	apple-store	https://sta		
10530123	Clash Roy	Supercell	Strategy	2016-01-0	2024-11-1	764	24.77066	237	73.1099	50	22.0479	19	4.8422	29	16.836	193	83.164	2024-06	apple-store	https://sta		
10530123	Clash Roy	Supercell	Strategy	2016-01-0	2024-11-1	764	24.87836	237	72.9362	49	21.7632	20	5.3006	30	17.4943	191	82.5057	2024-07	apple-store	https://sta		
10530123	Clash Roy	Supercell	Strategy	2016-01-0	2024-11-1	764	24.67572	240	73.8904	48	21.1661	19	4.9435	32	18.3404	189	81.6596	2024-08	apple-store	https://sta		
10530123	Clash Roy	Supercell	Strategy	2016-01-0	2024-11-1	764	24.40523	244	75.0869	46	20.5085	17	4.4046	31	17.8594	190	82.1406	2024-09	apple-store	https://sta		
10530123	Clash Roy	Supercell	Strategy	2016-01-0	2024-11-1	764	24.41537	243	75.0312	46	20.5527	17	4.4161	32	18.2681	189	81.7119	2024-10	apple-store	https://sta		
52947919	Clash of C	Supercell	Strategy	2012-06-1	2024-11-0	392	24.86606	324	72.0047	53	23.5133	17	4.482	25	14.375	198	85.625	2023-12	apple-store	https://sta		
52947919	Clash of C	Supercell	Strategy	2012-06-1	2024-11-0	392	24.87598	234	72.0654	53	23.3487	18	4.5859	24	14.1825	199	85.8175	2024-01	apple-store	https://sta		
52947919	Clash of C	Supercell	Strategy	2012-06-1	2024-11-0	392	25.03126	231	71.1371	55	24.1621	18	4.7008	26	14.8989	197	85.1011	2024-02	apple-store	https://sta		
52947919	Clash of C	Supercell	Strategy	2012-06-1	2024-11-0	392	25.10557	229	70.5997	56	24.7199	18	4.6805	27	15.6516	196	84.3484	2024-03	apple-store	https://sta		
52947919	Clash of C	Supercell	Strategy	2012-06-1	2024-11-0	392	25.06198	230	71.0405	55	24.1659	18	4.7936	27	15.5888	196	84.4112	2024-04	apple-store	https://sta		
52947919	Clash of C	Supercell	Strategy	2012-06-1	2024-11-0	392	25.32067	226	69.7974	57	24.9732	20	5.2394	26	15.3707	196	84.6293	2024-05	apple-store	https://sta		
52947919	Clash of C	Supercell	Strategy	2012-06-1	2024-11-0	392	25.48033	223	68.7581	59	25.9627	20	5.2792	26	14.8195	197	85.1805	2024-06	apple-store	https://sta		
52947919	Clash of C	Supercell	Strategy	2012-06-1	2024-11-0	392	25.61821	221	68.2118	59	26.1831	22	5.6051	25	14.7951	197	85.2049	2024-07	apple-store	https://sta		
52947919	Clash of C	Supercell	Strategy	2012-06-1	2024-11-0	392	25.33521	225	69.4619	58	25.4982	19	5.0399	26	15.2505	196	84.7495	2024-08	apple-store	https://sta		
52947919	Clash of C	Supercell	Strategy	2012-06-1	2024-11-0	392	25.40019	224	68.993	59	25.9838	19	5.0232	26	15.1217	197	84.8783	2024-09	apple-store	https://sta		
52947919	Clash of C	Supercell	Strategy	2012-06-1	2024-11-0	392	25.43898	223	68.9014	59	25.9337	20	5.1649	27	15.6124	196	84.3876	2024-10	apple-store	https://sta		
40688913	Coin Mast	Moon Act Party	Luck	2010-12-1	2024-11-1	267	35.2809	73	22.6112	125	54.8359	88	22.5529	112	64.2432	83	35.7568	2023-12	apple-store	https://sta		
40688913	Coin Mast	Moon Act Party	Luck	2010-12-1	2024-11-1	267	35.22807	73	22.5685	126	55.2068	87	22.2248	110	62.9761	86	37.0239	2024-01	apple-store	https://sta		
40688913	Coin Mast	Moon Act Party	Luck	2010-12-1	2024-11-1	267	35.25628	77	23.7808	120	52.8609	91	23.3583	109	62.2358	87	37.7642	2024-02	apple-store	https://sta		
40688913	Coin Mast	Moon Act Party	Luck	2010-12-1	2024-11-1	267	35.18727	76	23.4858	122	53.7769	89	22.7373	108	62.0101	88	37.9899	2024-03	apple-store	https://sta		

8) 게임별 앱 아이콘 추출

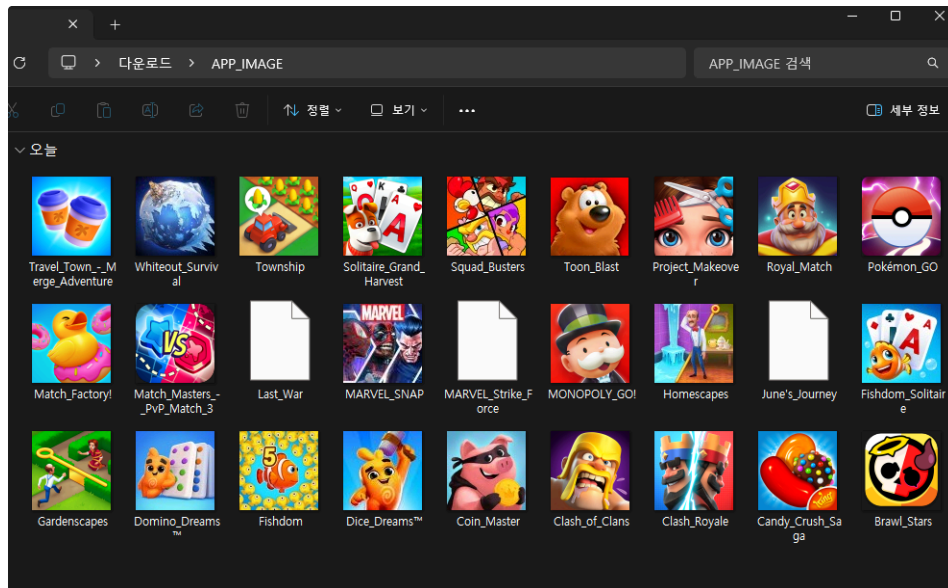
8

```

1 # 이미지파일 추출 함수 생성
2 def download_images(df, folder_path="내 컴퓨터의 저장공간 위치 입력"):
3
4     if not os.path.exists(folder_path):
5         os.makedirs(folder_path)
6
7     for _, row in df.iterrows():
8         product_name = row["product_name"]
9         icon_url = row["icon_web"]
10        file_name = f"{product_name.replace(' ', '_').replace('/', '_')}.png"
11        file_path = os.path.join(folder_path, file_name)
12
13        try:
14            response = requests.get(icon_url, stream=True)
15            if response.status_code == 200:
16                with open(file_path, "wb") as file:
17                    file.write(response.content)
18                print(f"Downloaded: {file_name}")
19            else:
20                print(f"Failed to download {file_name}: HTTP {response.status_code}")
21        except Exception as e:
22            print(f"Error downloading {file_name}: {e}")
23
24
25 # 이미지파일 추출
26 download_images(demographics_google[['product_name', 'icon_web']].drop_duplicates(ignore_index= True))

```

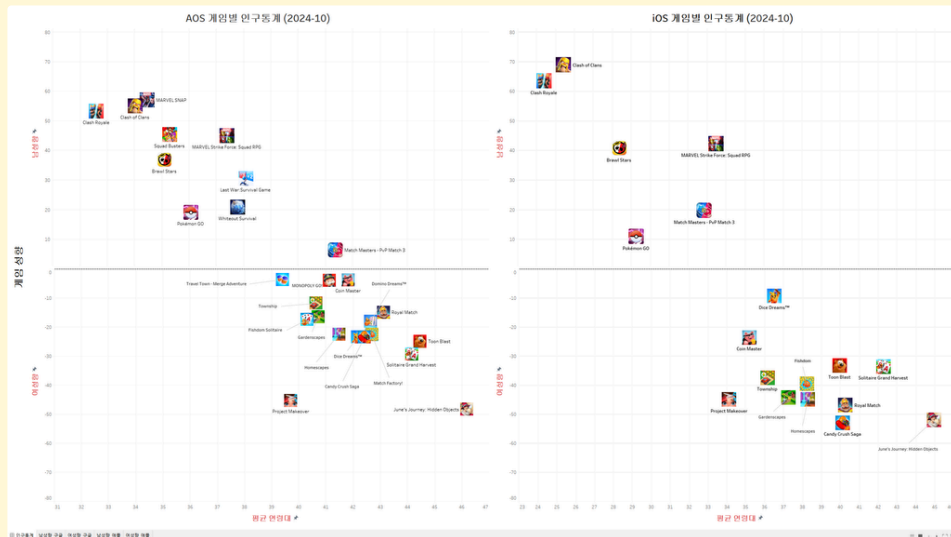
결과 예시



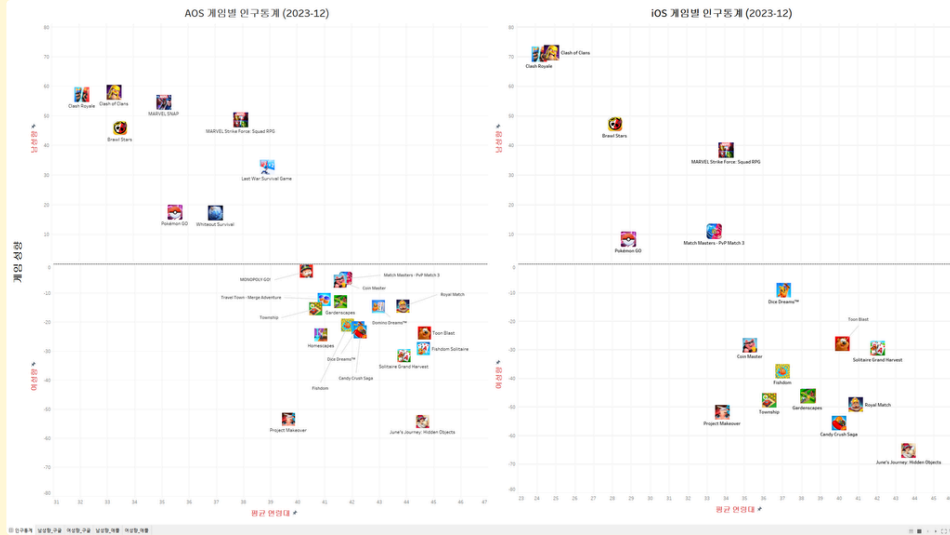
3. 시각화

대시보드

- 2024-10 기준 게임별 인구통계 데이터 대시보드



- 2023-12 기준 게임별 인구통계 데이터 대시보드



📍대시보드 관련 설명

• 대시보드 축 관련

- 평균 연령대 : 각 게임 유저들의 연령별 분포를 고려하여 추정 평균 연령대 생성
- 게임성향 - 남성향 : (남성유저 비율 - 여성유저 비율) = 양수인 경우,
 - ---> y축 위쪽에 형성 & y축과 멀수록 남성향
- 게임성향 - 여성향 : (남성유저 비율 - 여성유저 비율) = 음수인 경우,
 - ---> y축 아래쪽에 형성 & y축과 멀수록 여성향

• 대시보드 필터 관련

- 'Date' 필터로 원하는 달의 데이터 선택 가능 (다중선택 O)
- 'Product_Name' 필터로 원하는 게임만 시각화 가능 (다중선택 O)

• 예시 : AOS 게임별 인구통계 (2024-10) 그래프

- Clash Royale : (남성유저 비율 - 여성유저 비율) = 53.2%, 대부분 남성 유저로 구성되어 있음
- June's Journey : (남성유저 비율 - 여성유저 비율) = -48.3%, 대부분 여성 유저로 구성되어 있음